

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CO3133 – Deep Learning and Its Applications

Assignment 2 – Semester 261

Dataset & Project Proposal

Semantic Segmentation on Cityscapes

Advisor(s): Dr. Lê Thành Sách

Student(s): **Group RTX5090**

Đặng Thành Phú 2452967

Cao Đức Thành 2453164

HO CHI MINH CITY, OCTOBER 2026



Contents

1	Dataset Information	4
2	Task, Input, and Output Formulation	4
3	Data Distribution and Preliminary EDA	5
4	Data Splitting and Leakage Prevention	6
4.1	Proposed experimental split	6
4.2	Leakage control and reproducibility	7
5	Evaluation Metrics	7
6	Planned Methods and Architectures	8
7	Compute Resources and Feasibility	9



1 Dataset Information

This project uses the **Cityscapes Dataset**, obtained from the official Cityscapes website at <https://www.cityscapes-dataset.com/>. The dataset provides high-resolution urban street-scene images together with dense pixel-level annotations, making it suitable for semantic segmentation.

The two required packages are:

- 1) **Fine Annotations** — `gtFine_trainvaltest.zip` (approximately 241 MB).
Download: <https://www.cityscapes-dataset.com/file-handling/?packageID=1>.
This package contains semantic label masks, instance-ID masks, colored label visualizations, and polygon annotations. Fine annotations are publicly available for 2,975 training images and 500 validation images. The labels of the 1,525 test images are withheld for benchmark evaluation.
- 2) **RGB Images** — `leftImg8bit_trainvaltest.zip` (approximately 11 GB).
Download: <https://www.cityscapes-dataset.com/file-handling/?packageID=3>.
This package contains 5,000 PNG street-scene images at an original resolution of 2048×1024 pixels (width $\times$ height). The official split consists of 2,975 training, 500 validation, and 1,525 test images.

Both packages are distributed under the **Cityscapes License**.

2 Task, Input, and Output Formulation

The objective is **urban street-scene semantic segmentation**: every pixel of an input image is assigned to one semantic category.

- **Input:** an RGB image

$$X \in \mathbb{R}^{H \times W \times 3},$$

originally at 2048×1024 pixels. Images may be resized or cropped during training.

- **Output:** a pixel-wise segmentation mask

$$\hat{Y} \in \{0, 1, \dots, 18\}^{H \times W},$$

where each value identifies the predicted semantic class of the corresponding pixel.



- **Target classes:** 19 classes are used from the 34 annotated Cityscapes classes: road, sidewalk, building, wall, fence, pole, traffic light, traffic sign, vegetation, terrain, sky, person, rider, car, truck, bus, train, motorcycle, and bicycle.
- **Ignored pixels:** ground-truth pixels outside the 19 target classes are assigned the ignore value 255. They are excluded from loss computation and evaluation and are not treated as an additional predicted class.

3 Data Distribution and Preliminary EDA

Pixel-level exploratory analysis was conducted on all 2,975 images in the official training partition. Of the 6,239,027,200 total pixels, 5,523,279,889 are valid pixels belonging to the 19 evaluation classes, while 715,747,311 pixels are ignored. Ignored pixels therefore account for 11.47% of all training pixels. The class distribution among valid pixels is reported in Table 3.1; the image count indicates the number of training images in which each class occurs.

Table 3.1: Pixel and image frequency of the 19 evaluation classes in the official training partition.

Class	Pixels	Valid pixels (%)	Images
Road	2,036,416,525	36.869696	2,934
Building	1,260,636,120	22.824049	2,934
Vegetation	879,783,988	15.928651	2,891
Car	386,328,286	6.994545	2,832
Sidewalk	336,090,793	6.084986	2,811
Sky	221,979,646	4.018982	2,686
Pole	67,789,506	1.227341	2,949
Person	67,326,424	1.218957	2,343
Terrain	63,949,536	1.157818	1,654
Fence	48,454,166	0.877272	1,296
Wall	36,199,498	0.655399	970
Traffic sign	30,448,193	0.551270	2,808
Bicycle	22,861,233	0.413907	1,646

Continued on next page

Class	Pixels	Valid pixels (%)	Images
Truck	14,772,328	0.267456	359
Bus	12,990,290	0.235192	274
Train	12,863,955	0.232904	142
Traffic light	11,477,088	0.207795	1,658
Rider	7,463,162	0.135122	1,023
Motorcycle	5,449,152	0.098658	513

The distribution is highly imbalanced: road, building, and vegetation together account for 75.62% of valid pixels, whereas each of the ten least represented classes contributes less than 1%. Motorcycle is the rarest class by pixel count (0.098658%), while train occurs in the fewest images (142 of 2,975). Consequently, the main anticipated challenges are reliable recognition of rare classes and accurate boundary delineation between adjacent objects.

4 Data Splitting and Leakage Prevention

4.1 Proposed experimental split

The official partitions will be reorganized only where labels are available, producing the split in Table 4.1. Specifically, 500 images from the official training partition are reserved for validation, while the 500-image official labeled validation partition is used as a local test benchmark.

Table 4.1: Proposed use of the 5,000 Cityscapes images.

Partition	Images	Percentage	Purpose
Train	2,475	49.5%	Model optimization
Validation	500	10.0%	Hyperparameter tuning and check-point selection
Local test	500	10.0%	Labeled benchmark for mIoU and Dice evaluation
Unlabeled test	1,525	30.5%	Qualitative inference only because labels are withheld
Total	5,000	100.0%	

4.2 Leakage control and reproducibility

The following controls will be applied:

- preserve the official training, validation, and test partitions and prevent overlap between them;
- respect the Cityscapes city/sequence-level partitioning;
- keep each RGB image paired with its corresponding segmentation mask;
- apply data augmentation exclusively to the training images; and
- use a fixed random seed of 42 for reproducibility.

To accommodate hardware limitations, images will be resized to 1024×512 pixels or randomly cropped into patches such as 512×512 during training.

5 Evaluation Metrics

The primary metric is **mean Intersection over Union (mIoU)**. For class c , IoU is defined as

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad (5.1)$$

and mIoU is the average over the 19 target classes. A higher mIoU indicates stronger overlap between the predicted and ground-truth masks.

Secondary metrics include:

- **Mean Dice score (F1-score):** segmentation similarity computed per class and averaged over all 19 classes;
- **Per-class IoU and Dice score:** detailed performance analysis, particularly for rare classes such as train, motorcycle, and rider; and
- **Overall pixel accuracy:** the proportion of correctly classified pixels among all valid, non-ignored pixels.

6 Planned Methods and Architectures

Two principal model families will be compared under the same data and evaluation settings:

1. **Baseline: U-Net trained from scratch.** U-Net uses an encoder to extract image features and a decoder to recover spatial resolution. Skip connections transfer detailed features between corresponding stages, helping the model preserve object boundaries. All parameters will be randomly initialized.
2. **Pretrained model: DeepLabV3 with a ResNet-50 backbone pretrained on ImageNet.** DeepLabV3 uses atrous convolutions and Atrous Spatial Pyramid Pooling to capture context at multiple scales. The pretrained backbone provides initial visual features, while a randomly initialized segmentation head will learn to predict the 19 Cityscapes classes.

For DeepLabV3, the ResNet-50 backbone will initially be frozen while the segmentation head is trained. The backbone will then be unfrozen and the full network fine-tuned using a lower learning rate for its pretrained layers. Results will be compared with U-Net using identical training and validation data.



7 Compute Resources and Feasibility

Table 7.1: Estimated computational requirements.

Item	Estimate
Hardware	Local GPU or Google Colab; preferably an NVIDIA T4 with 16 GB VRAM or equivalent
Dataset	5,000 RGB images: 2,975 official training, 500 validation, and 1,525 test images
Disk storage	Approximately 25–40 GB for compressed/extracted data, checkpoints, and outputs
Input resolution	1024×512 pixels, with optional 512×512 training crops
Batch size	Approximately 2–4 images, adjusted according to available GPU memory
Training schedule	Approximately 30–50 epochs with mixed-precision training

The proposed setup is feasible because only 2,475 images are used for model fitting, while 500 images are reserved for validation monitoring and the 500 official validation images form the local benchmark. Resizing, random cropping, mixed precision, and an adjustable batch size will reduce memory demand. Actual training time and GPU memory consumption will be measured during implementation.